Week 5 Problem Set: Testing and Classifying Continuity

Calculus Honors · Module 3: Continuity & the Great Theorems · Lessons 3.01 and 3.02

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Keep your writing neat and vertical, line up your equal signs, and label your final answers. Check all three conditions in order, and name each discontinuity by type.

PART A · TESTING THE THREE CONDITIONS

1. Test whether $f(x) = \dfrac{x^2 - 9}{x - 3}$ is continuous at $x = 3$. Check all three conditions in order and state which one fails, if any.

2. Let $g(x) = \begin{cases} x + 2 & x \neq 4 \\ 6 & x = 4 \end{cases}$. Test whether g is continuous at $x = 4$, checking all three conditions.

PART B · CONTINUITY ON AN INTERVAL

3. Find every x-value where $h(x) = \dfrac{1}{x^2 - 4}$ fails to be continuous, and justify each one using the three-part definition.

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4. Explain, using the limit laws from Lesson 1.02, why $f(x) = 3x^4 - 2x + 7$ is continuous at every real number.

PART C · CLASSIFYING DISCONTINUITIES

5. Classify the discontinuity of $p(x) = \dfrac{x^2 - 1}{x + 1}$ at $x = -1$.

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6. Let $q(x) = \begin{cases} x & x < 2 \\ x + 3 & x \geq 2 \end{cases}$. Classify the discontinuity of q at $x = 2$, stating both one-sided limits.

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7. Classify the discontinuity of $r(x) = \dfrac{1}{x - 5}$ at $x = 5$, stating both one-sided limits.

8. A function is identical to $y = 3 - x^2$ everywhere except that it has an open circle at $(0, 3)$, with no point plotted there at all. Classify the discontinuity at $x = 0$, and explain how you would repair it.

PART D · EXPLAIN YOUR THINKING

9. Convince a classmate, in three or four sentences, why a jump discontinuity cannot be repaired by redefining a single value, the way a removable discontinuity can. Use one of the examples above as evidence.

10. Describe, in words or with a sketch, a function that is continuous everywhere except for one infinite discontinuity. State where the discontinuity occurs and how you know it is infinite rather than removable or jump.